TOOLPLANE / ERPBRIDGE

ERPBridge Evaluation Report

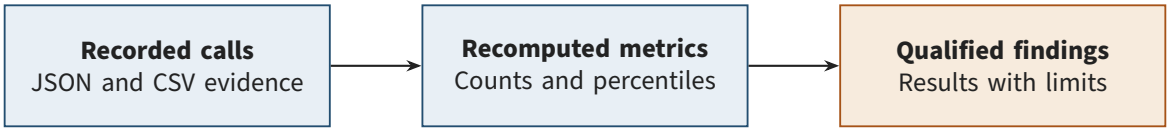
Integration evidence, read performance, governance checks, and evaluation limits

Evidence review edition | 19 October 2026

100/100
Demo read calls
One short burst workload

361 ms
Median latency
Client-observed, nearest rank

662 ms
p95 latency
Client-observed, nearest rank



What the evidence supports
Integration feasibility, selected boundary behavior, and a bounded read-latency observation.

Interpretation limit
This is not a release certification, a maximum-capacity benchmark, or proof of production reliability. Historical run-to-binary attribution is unknown; the separate demo target version was not recorded.

Prepared from the supplied evidence archive. No new live benchmark or model trial was run to produce this report.

01 / DECISION SUMMARY

What the available evidence supports

Conclusion. ERPBridge has retained evidence of ERP integration feasibility and selected governance behavior. A separate demo run adds client-observed read latency for 100 calls. These findings are useful for a presentation, but they do not establish complete conformance or production readiness.

ToolPlane is the proposed architecture for exposing a legacy enterprise resource planning (ERP) application programming interface (API) through Model Context Protocol (MCP) tools. ERPBridge is the reference system. OpenAPI describes Hypertext Transfer Protocol (HTTP) operations; generated tool resources connect the agent-facing interface to those operations. This report evaluates the supplied records, not every implementation property.

Table 1. Evidence overview and permitted interpretation

Evidence	Recorded result	Supported conclusion
Demo reads	100/100 calls; 361 ms median; 662 ms p95	Bounded read completion and client-latency observation
Historical integration	10 sequence records; 50 status-200 outcomes	Feasibility for five ERPNext operations
Structural conversion	7/8 rows pass all structural checks	Preliminary transformation fidelity; one documented limitation
Access decisions	3/3 cases match expectations	One allowed and two denied role cases
Boundary checks	Seven scenario groups, ten retained records each	Selected behavior, not full invariant validation
Missing evaluations	Zero model-trial rows; 24 blocked evolution rows	No measured usability or maintenance benefit

Source: S1–S10 and S14; source register at the end of this report.

Keep the evidence sets separate
The 100-call demo run is not an extension of the 50-outcome historical integration set. Its target version, cache/auth configuration, and hardware profile are not established. Do not pool counts as a single 150-call benchmark.

How to read this report

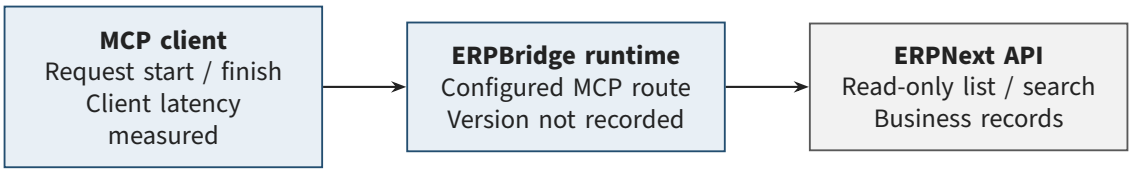
The next pages cover the workload, latency results, historical transformation and governance evidence, exploratory overhead, documentation-based positioning, provenance, and next evaluation steps. Vector charts are computed from retained samples or clearly labelled summary records. No new live workload or paid model trial was run for this document.

Interpretation limit
Historical run-to-binary attribution is unknown. Success counts describe retained observations, not guarantees. Copy integrity, numerical consistency, semantic correctness, and release integrity are different checks.

02 / METHOD

Read workload and measurement boundary

The demo report records a read-only run started at `2026-10-18T17:47:18.724Z` through the configured `mcp__erpnext_bridge` connection. Figure 1 shows the measurement boundary at a conceptual level; it is not a trace of internal processing.



Timing covers the client-observed call, not isolated runtime processing.
Server CPU, memory, database load, and other-user impact were not measured.

Figure 1. Client-observed timing includes the call path. Internal stage timings were not collected.

Source: S1 and S2; conceptual diagram derived from the recorded connection description.

Table 2. Demo workload conditions

Condition	Recorded setting
Calls and batching	100 calls; 10 batches of 10 concurrent calls
Pacing	Batch starts at least one second apart, as stated in the source report
Operation mix	Five read tools, 20 calls each
Projection and page cap	Only `name`; maximum 100 rows per call
Writes / retries	None, as stated in the source report
Logging	Status, duration, batch, row count, and error; no response-body logging
Stop rule	Stop after five failed calls; actual recorded stop is null
Deployment identity	Target version and server resource metrics unavailable

Definitions used in the charts

Latency is the recorded client duration in milliseconds. Percentiles use nearest rank: for sorted observations $x_{(1)}, \ldots, x_{(n)}$, $P_p = x_{(\lceil pn \rceil)}$. Thus the median for 100 calls is the 50th ordered duration, not the average of the 50th and 51st.

Completion rate equals $100/(11946/1000) = 8.3710$ calls/s. Its denominator includes pacing and connection setup. It is not the configured offered rate or a measured maximum capacity.

Interpretation limit
The JSON has batch indices but no per-call or batch-start timestamps. Counts and percentiles can be recomputed. Concurrent overlap and the reported pacing cannot be reconstructed independently from these fields.

03 / BENCHMARK OBSERVATIONS

Overall read latency

All 100 raw samples have `ok=true` and `error=null`. The supplied values reproduce under nearest-rank calculation. Figure 2 uses the complete ordered sample, rather than a fitted distribution.

100/100
Completed successfully
No recorded tool errors

11.946 s
Total wall time
Includes pacing

8.37/s
Completion rate
Not a capacity estimate



Figure 2. Empirical cumulative distribution of 100 client-observed read durations. The x-axis starts at zero; the shortest observed call was 205 ms.

Source: S2; recomputed by `analyze_evidence.py`.

Table 3. Overall latency summary

Metric	Median	p95	p99	Maximum
Client latency, ms	361	662	691	695
Ordered rank / 100	50	95	99	100

Interpretation

The run demonstrates completion of this short, small-result read workload. Tail latency is materially above the median within the sample. No latency service-level objective was supplied, so these values are observations rather than a pass/fail against a service target.

Interpretation limit
A 100/100 observed success count is not a population reliability estimate or production guarantee. There is no direct-HTTP control in this run. These values cannot isolate bridge overhead, server processing time, network delay, or database time.

04 / BENCHMARK DETAIL

Per-tool latency and returned rows

Every tool has 20 successful samples. Figure 3 compares sample medians and p95 values. Tool names are shortened on the axis; Table 4 gives the exact names.

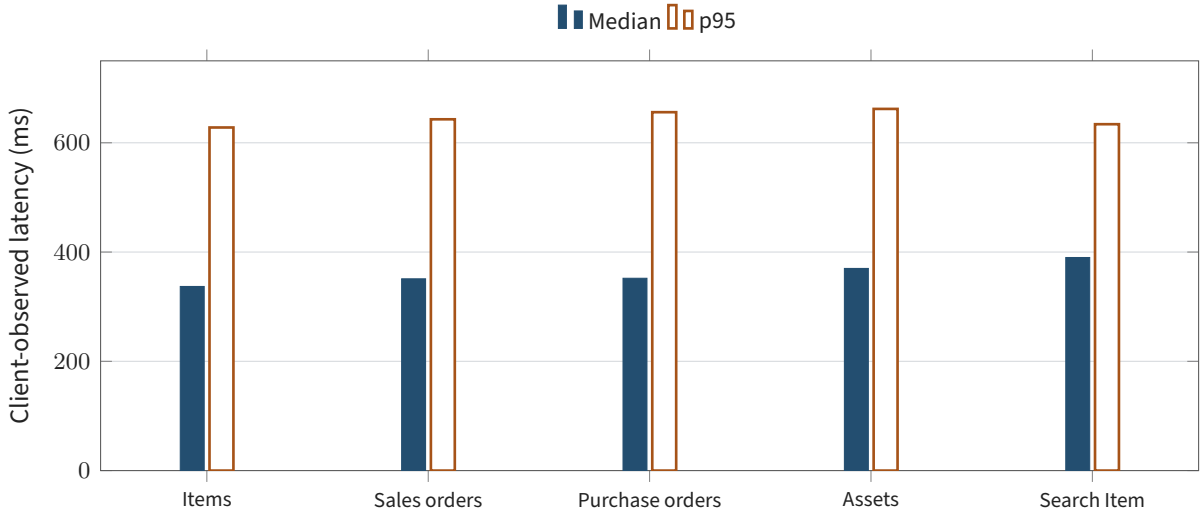


Figure 3. Per-tool client latency. Filled bars are medians; outlined bars are p95. Bars share a zero baseline.

Source: S2; nearest rank, 20 calls per tool.

Table 4. Per-tool results with exact operation names

Tool	Calls	Rows / response	Median ms	p95 ms
`list_items`	20/20	56	337	628
`list_sales_orders`	20/20	5	351	643
`list_purchase_orders`	20/20	10	352	656
`list_assets`	20/20	3	370	662
`search_records(Item)`	20/20	56	390	634

What this comparison does and does not show

The observed medians range from 337 to 390 ms and p95 values from 628 to 662 ms. Collections differ in size and the calls share the same burst run. These summaries do not support a causal claim that one tool implementation is faster or a statistically established ranking of tool performance.

The page cap was 100, but the largest collection returned 56 rows. Only `name` was requested. This workload did not stress large-result serialization, broad record projection, pagination at the cap, writes, or mixed business transactions.

Use the correct unit
The rows column is the number of records per response, not the number of requests or an independent correctness score. Row counts and success flags do not validate the business meaning of every returned record.

05 / RUN VARIATION

Batch-level latency distribution

The raw evidence retains batch membership. Figure 4 shows each batch's nearest-rank median and minimum–maximum range. It deliberately uses a batch index, not a synthetic wall-clock timeline.

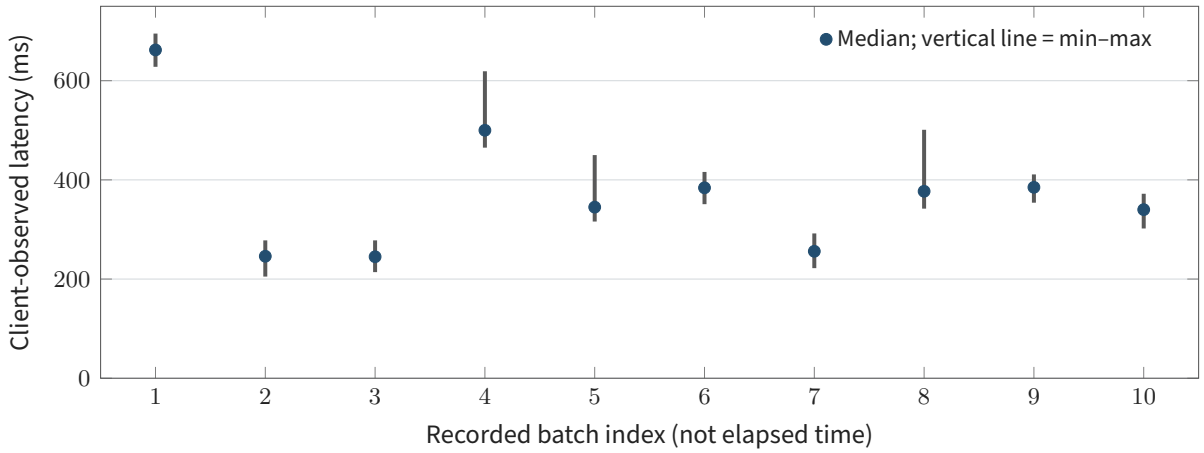


Figure 4. Recorded latency range and median for each ten-call batch. The first batch includes connection warm-up according to the source report.

Source: S1 and S2; batch summaries recomputed from raw samples.

Observed variation

The first batch spans 628–695 ms, with a median of 662 ms. Batches 2 and 3 have medians of 246 and 245 ms. Batch 4 rises to 500 ms. These differences show variation within the short run; the available fields do not establish its causes.

The source report states that the first batch includes connection warm-up. All 100 samples remain in the overall analysis. Removing warm-up observations after viewing the results would define a different analysis and must be labelled separately.

Table 5. Interpretation of workload and rate terms

Term	Meaning in this report
Burst batch size	Ten calls launched concurrently, as reported by the source
Batch interval	Reported start pacing of at least one second
Completion rate	100 completions divided by 11.946 seconds of total wall time
Maximum sustainable rate	Not measured; no saturation search or steady-state campaign
Resource efficiency	Not measured; CPU, memory, and database metrics absent

Interpretation limit
Do not present this as an evenly spaced 10-RPS stream or a sustained 10-RPS capacity test. The JSON cannot verify overlap, queue depth, server utilization, or latency experienced by other users.

06 / HISTORICAL EVIDENCE

Integration and transformation fidelity

The retained happy-path directories contain ten invocation records, each with five status-200 operations: `list_companies`, `list_items`, `get_item`, `create_item`, and `delete_item`. The source report also describes direct Item creation/deletion state checks. This supports integration feasibility in an earlier ERPNext profile, not unchanged ERP code or general reliability. [S3, S4]

A separate eight-row structural probe covers six MockERP operations and two sanitized ERPNext operations. Figure 5 shows that seven rows passed all checks. All eight valid-call statuses are `unavailable`; structural fidelity is not a live invocation result. [S5]

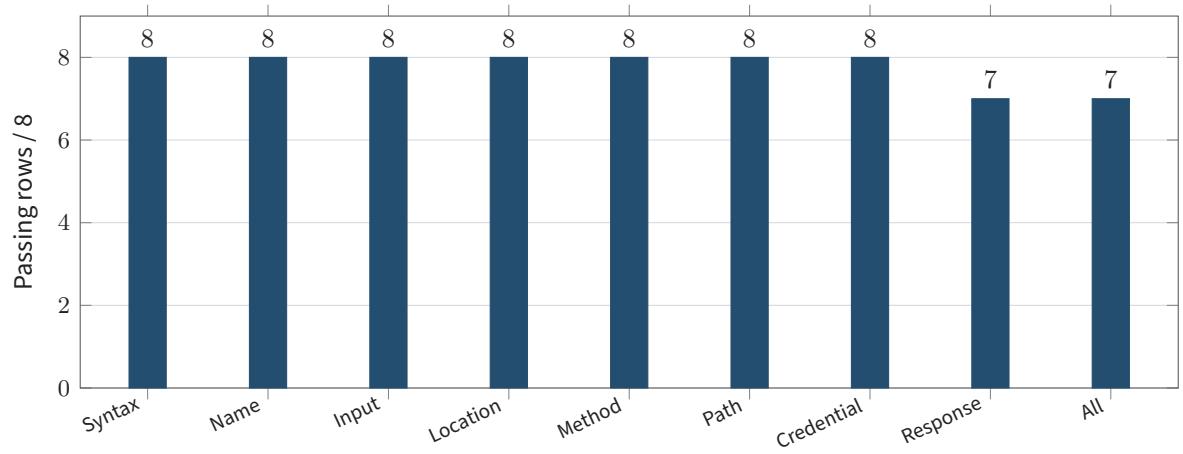


Figure 5. Historical structural-check counts. Every bar uses the same eight-row denominator; the failed row is retained.

Source: S5; row-level CSV, not a general OpenAPI coverage benchmark.

The response-mapping failure

The failing operation is `GET /integration/echo/last`. Its expected `data` object uses `allOf` and `nullable`, outside the declared historical subset. The generator emitted `responsePath: data` without an output schema. The profile audit calls this an **eligibility-oracle inconsistency and response-mapping limitation**. It is neither a supported-pass nor a corrected rerun. [S6]

Preserve the denominator
State "7 of 8 historical probe rows passed all structural checks." Do not turn this into a 7/7 supported-subset claim or general 87.5% API-conversion accuracy.

Interpretation limit
The integration records are labelled smoke/non-final and use repetition identifier 1. Dispatch totals were assigned, not independently counted. The before/after ERP baseline hashes are placeholders, not measurements of unchanged ERP code, contracts, or deployment. [S3, S4, S10]

07 / GOVERNANCE OBSERVATIONS

Access decisions and boundary checks

Three governance CSV rows match their expected outcomes: one allowed call, one wrong-role denial, and one missing-role denial. All record HTTP 200; `is_error=true` distinguishes the two denied results. HTTP status alone is not the authorization decision. The collection note says the existing CSVs lacked raw provenance during that workspace check. [S7]

Table 6. Selected historical boundary observations and gaps

Boundary	Retained observation	Restriction
Admission	Changed same-version content rejected in 10/10 records; digest unchanged	Generic HTTP 500 does not identify the exact rejection rule or reviewer authority
Convergence	Controller reports convergence after failure/restart in 10/10 records	No retained recovery duration, repaired discovery, or later ERP call
Authorization	Denied calls record zero connector entries in 10/10 records	One fixed-principal denial path; not broad policy coverage
Queued withdrawal	Instrumented queued calls record zero connector entries in 10/10 records	Pause arrival unacknowledged; not proof of distributed revocation
Revision checks	Serving transition, serving-revision digest binding, and mutated-digest rejection	No active non-serving witness or accepted call during transition
Credential reference	Opaque reference present; authenticated calls succeed	No comprehensive secret-leakage or credential-lifecycle evaluation

Source: S3, S8, and S9; specific restrictions follow the retained report.

What “observed” means

The conformance summary marks six invariants `observed`, but also states `partial` architecture coverage, `unknown` release integrity and historical attribution, and `not-release-verified` overall status. A six-of-six green pass chart would erase this distinction.

Seven scenario groups have ten retained boundary records each. Some groups support multiple requirements. Their counts do not establish 70 independent proofs, unique repetitions, complete scenario coverage, or behavior of a later implementation.

Interpretation limit
The earlier boundary profile used behavior-affecting instrumentation. These observations must not be transferred automatically to current-source commitment, authority, or revision-selection mechanisms. Admission records also do not independently prove prior human review or a separate approval authority. [S3, S8–S10]

08 / EXPLORATORY COMPARISONS

Overhead and discovery consistency

Historical direct-HTTP comparison

Two CSV summary rows concern `GET /resource/Company`. Each is labelled 30 runs. They are not the same workload, deployment, or timing regime as the 100-call demo read test. [S11]

Table 7. Exploratory latency summaries, milliseconds

Configuration	Direct HTTP	ToolPlane	Difference
Auth off, cache off: p50	20.352	14.498	—
Auth off, cache off: p95	23.115	24.738	+1.623
Admin auth, cache off: p50	21.020	15.355	—
Admin auth, cache off: p95	24.005	23.664	−0.341

Source: S11; differences shown only where recorded, p95 = ToolPlane minus direct.

The negative p95 difference does not show that ERPBridge accelerates ERP access. These are sparse aggregate rows without the trial-level timing traces needed for uncertainty analysis or a general overhead bound. Configuration and provenance limits prevent a direct comparison with the hundreds-of-milliseconds demo results.

Discovery/version consistency case

Two descriptive rows record reverse registration order, `v2_then_v1`. Before the fix, discovery omitted a v2-only argument although resolution selected version 2.0.0. After the fix, discovery included the argument and the recorded agreement flag became true. [S12]

Table 8. Recorded discovery/resolution agreement

Record	v2 argument shown	Resolved version	Agreement
Before fix	No	2.0.0	False
After fix	Yes	2.0.0	True

The pre-fix row records HTTP 200 for a v2-only-argument call. The post-fix invocation field is blank. A blank is missing evidence, not proof of a successful post-fix invocation. This is a useful defect-and-repair example, not a complete regression suite.

Interpretation limit
Do not pool separate experiments, convert sparse summaries into a capacity claim, or interpret absent post-fix measurements as passes. The report distinguishes a recorded consistency repair from complete per-call revision safety.

09 / ARCHITECTURE POSITIONING

Documentation-based capability comparison

The retained matrix compares eight criteria against pinned sources. Table 9 reproduces its ratings, not a new product evaluation. External references were not retrieved or independently revalidated for this PDF. [S13]

Table 9. Retained version-specific capability ratings

Criterion	AutoMCP v4 paper	FastMCP 4.0.3	Context Forge 1.0.9	ERPBridge prototype
API-derived tools	D	D	D	P
Selective exposure	D	D	D	P
Explicit revision targeting	NE	D	NE	P
Approval-bound exact resource	NE	NE	NE	P
Desired-state reconciliation	NE	NE	NE	P
Per-call revision consistency	NE	NE	NE	P
MCP client authentication	NE	D	D	P
Upstream credential mediation	D	D	D	P

Source: S13; AutoMCP source is arXiv:2507.16044v4, a preprint.

The legend carries the meaning
D: directly documented by the pinned source. **P**: partial or profile-specific implementation property. **NE**: not established by the reviewed source, not evidence of absence.

Architecture interpretation

The comparison positions ToolPlane around explicit tool resources, desired state, revision identity, and runtime governance rather than API conversion alone. This is a design-positioning argument derived from the archived matrix, not a measured superiority claim.

A resource can bind an input contract, HTTP mapping, and credential reference. An admission record does not, by itself, prove an independent reviewer or prior human approval. A stored desired-state model does not, by itself, prove measured recovery time. A version selector does not, by itself, prove that policy and execution share one immutable per-call revision.

Source and version caution

The prototype rows describe an evaluated profile. They must not be read as current-source certification or general guarantees across all ToolPlane deployments. The historical advanced behavior is not tied to every retained run by an observed immutable binary identifier.

Interpretation limit
Do not count D/P/NE cells as performance scores or claim that NE systems lack a feature. This table does not benchmark latency, task success, reliability, maintenance effort, or security quality.

10 / EVIDENCE INTEGRITY

Provenance, archive state, and limitations

The historical audit distinguishes retained behavior from release attribution. Placeholder source/image identifiers and ERP baseline hashes, an unauthorized freeze, incomplete frozen coverage, and author-authored expected results prevent a release-level conformance claim. They do not prove that the recorded calls never occurred. [S10]

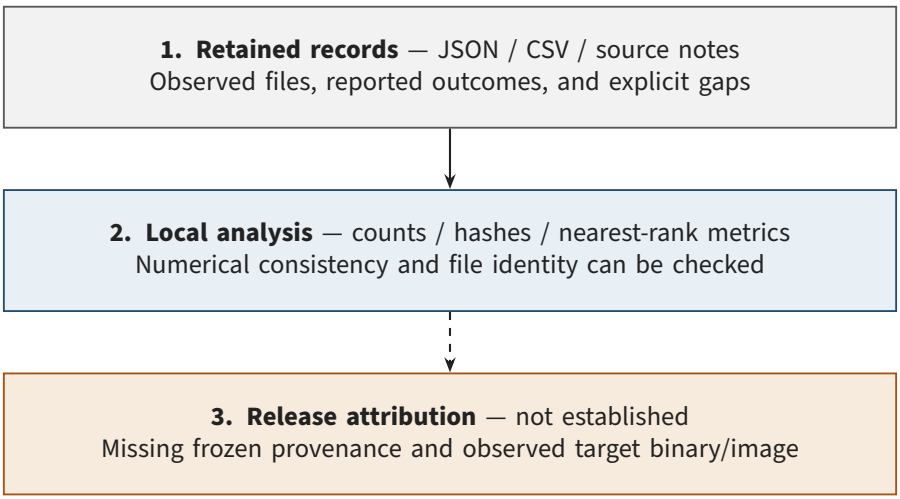


Figure 6. Numerical verification does not close the release-attribution gap. The dashed step marks an unsupported inference, not a successful release gate.

Source: S10 and the report's local analysis; editorial evidence model.

Archive state after the move

The original manifest lists 116 files. At report generation, 113 entries resolve at their listed paths with matching hashes. Two demo files have moved out of `.agents/artifacts/` into `sources/erpbridge-demo/`; both relocated files match their original manifest hashes. The copied `sources/paper/sections/evaluation.tex` is absent. There are therefore 115 source files available, not 116 complete manifest-path matches.

The Markdown report still contains links to the old demo paths and missing manuscript section. This PDF uses the current demo paths and identifies the Markdown report as a secondary summary where appropriate. The original files and manifest are preserved unchanged; the generated analysis records the discrepancies.

Principal validity limits

One historical single-process instrumented profile, one short demo read run, small collections, fixed principals, an author-authored oracle, missing target identity, and absent resource measurements limit generalization. There are no observed model-usability trials or completed evolution-repair measurements.

Interpretation limit
Neither matching file hashes nor recomputed percentiles establish business-semantic correctness, unchanged ERP deployment, independent review, server resource efficiency, or production readiness.

11 / RECOMMENDATIONS

Next evaluation and presentation guidance

A measured follow-up campaign

The following is a proposed evaluation plan, not work completed for this report. Establish acceptance thresholds before collecting results.

Table 10. Evidence required before stronger claims

Claim area	Required evidence
Release conformance	Authorized pre-run freeze, observed source/image/schema/SDK identifiers, independent oracle review, and complete scenario registry
ERP preservation	Before/after measurements of source, contracts, and deployment state; no synthetic baseline hashes
Sustained capacity	Controlled offered rates, warm-up policy, steady-state durations, unique repetitions, saturation search, error rates, and resource telemetry
Bridge overhead	Paired direct-HTTP and MCP measurements on the same target, configuration, payload, and time basis; raw timings and uncertainty analysis
Governance and recovery	Direct connector/HTTP instrumentation, broader identities and cache paths, synchronized withdrawal witnesses, timed recovery and post-recovery calls
Agent effectiveness	Frozen task set, fixed model/configuration, raw/generated/reviewed/manual conditions, recorded task success, and model-use provenance
Maintenance effort	Explicit mutation/repair/replay commands, repair time, changed lines, manual actions, and restored task success

Source: S3, S6, S10, S14; recommendations are design claims, not results.

Four defensible presentation statements

1. **Integration:** Ten historical sequence records report 50 successful outcomes across five ERP-Next operations.
2. **Transformation:** Seven of eight historical structural rows pass all checks, with a documented response-mapping/oracle limitation.
3. **Governance:** Three recorded role cases match expectations; selected boundary behavior is observed, not fully validated.
4. **Performance:** One read-only demo run completes 100 calls with 361 ms median and 662 ms p95 client latency under paced ten-call bursts.

Suggested presentation footer
Preliminary evidence. Historical run-to-binary attribution is unknown. Demo timings are client-observed and do not establish capacity, server-resource efficiency, or bridge overhead.

12 / TRACEABILITY

Local source register

All paths are relative to the `Audit Report` folder containing the delivered PDF. **R** abbreviates `sources/research/evidence/evaluation/`; **D** abbreviates `sources/erpbridge-demo/`. This is a local evidence register, not a newly conducted literature review.

S1 **Demo performance narrative.** `Derpbridge-perf-report.md`. Workload, connection, stop rule, reported percentiles, and limits.

S2 **Raw demo samples.** `Derpbridge-perf-evidence.json`. 100 per-call records, elapsed time, requested fields, page cap, and batch labels. Source for all demo charts.

S3 **Presentation evidence summary.** `REPORT.md`. Interpretations of historical integration and boundary observations. Some original links are stale after relocation; this PDF does not claim they all resolve.

S4 **Historical integration records.** `Rresults/task11/happy-path/`. Ten directories with invocation, metadata, baseline, state, dispatch, cleanup, and control/discovery records. Baseline placeholders are not preservation measurements.

S5 **Historical structural probe.** `Rresults/transformation.csv` and `results/transformation-failures.md`. Eight rows; response-mapping failure remains visible.

S6 **Transformation profile.** `Rtransformation-profile.md` and `openapi-support-matrix.md`. Historical eligibility audit and stricter requirements; not completed implementation coverage.

S7 **Governance.** `Rresults/governance.csv` and `results/governance-blockers.md`. Three descriptive decisions and raw-provenance/collection caveat.

S8 **Boundary observations.** `Rresults/task12/task12-final-06/boundary.json`. Retained combined scenario artifact, including instrumentation and limitations.

S9 **Conformance status.** `Rresults/invariant-conformance.json` and corresponding Markdown file. Partial coverage; not release-verified.

S10 **Historical provenance and release status.** `Rprovenance-audit.md`, `release-freeze.md`, and `archive-manifest.json`. Unknown historical execution identity; incomplete authorization.

S11 **Exploratory overhead.** `Rresults/overhead.csv`, with S7's collection caveat. Two summary rows for one operation.

S12 **Version consistency.** `Rresults/version-consistency.csv`. Before/after discovery agreement; post-fix invocation result absent.

S13 **Capability matrix.** `sources/research/evidence/comparison/system-capability-matrix.md`. Existing pinned-source ratings and external source links; not a quality benchmark.

S14 **Evaluation gaps and procedure.** `Rresults/agent-usability.csv`, `results/agent-usability.md`, `results/evolution.csv`, `results/repair-log.md`, and `protocol.md`. Empty model trials and 24 blocked evolution rows.

13 / REPRODUCIBILITY

How the report was produced

Analysis and build

The supplied records remain unchanged. The report project is in `pdf-report/`. Its Python analysis uses only the standard library and local evidence. It recomputes counts, nearest-rank percentiles, completion rate, per-tool summaries, batch ranges, and manifest-path/hash checks. It emits vector-plot coordinates and machine-readable results under `pdf-report/src/generated/`.

From the `Audit Report` folder:

```
python3 pdf-report/analyze_evidence.py
cd pdf-report
tectonic -X build --keep-logs -p
```

The canonical build target is `build/erpbridge-evaluation/erpbridge-evaluation.pdf`. The deliverable `ERPBridge-Evaluation-Report.pdf` is a copy of that output. The compiler log, QA report, and rendered pages are retained with the project.

Numerical and archive checks

The local analysis confirms 100 demo samples, 20 per tool, ten per batch, all success flags true, all error fields null, and the reported overall and per-tool percentiles. It also checks 50 status-200 historical operation outcomes, 7/8 structural passes, three matching access decisions, 24 blocked evolution rows, and zero usability data rows.

Archive checks identify relocation and absence rather than silently repairing history: 113 original manifest-path matches, two relocated demo files with matching original hashes, and one missing copied manuscript file. The generated `analysis-validation.json` records these details and SHA-256 values of key numerical inputs.

Report QA and scope

The document uses a restrained standalone report layout, selectable text, embedded fonts, and vector TikZ/PGFPlots figures. A design specimen was compiled and visually checked before the full document. Final structural checks and page renders are retained under `pdf-report/qa/`; see `pdf-report/QA-NOTES.md` for the final acceptance record.

No live ERP operation, model API call, new timing campaign, external literature discovery, or current-source implementation test was run to produce this PDF. Numerical consistency is not independent experimental replication. This report is not the IEEE conference manuscript and does not alter it.

Bottom line
Use the report to show measured observations with their exact denominators and limits. A new frozen, instrumented campaign is required before making release, capacity, reliability, maintenance, or model-effectiveness claims.